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A. Cut the Array

time limit per test: 2 seconds

memory limit per test: 512 megabytes

You are given an array of n non-negative integers $[a_1, a_2, \dots, a_n]$.

Your task is to cut it into three non-empty parts: a prefix, a middle part, and a suffix. Formally, you need to choose two integers l and r such that $1 \leq l < r < n$, and obtain three parts:

- the prefix up the element at index l inclusive (i.e., $[a_1, a_2, \dots, a_l]$);
- the central part from the element at index $l + 1$ to the element at index r inclusive (i.e., $[a_{l+1}, a_{l+2}, \dots, a_r]$);
- the suffix from the element at index $r + 1$ to n inclusive (i.e., $[a_{r+1}, a_{r+2}, \dots, a_n]$).

Let s_1, s_2, s_3 be the remainders of the sums of these parts modulo 3. In other words:

- $s_1 = \left(\sum_{i=1}^l a_i \right) \bmod 3$;
- $s_2 = \left(\sum_{i=l+1}^r a_i \right) \bmod 3$;
- $s_3 = \left(\sum_{i=r+1}^n a_i \right) \bmod 3$.

Your task is to find such boundaries l and r that either all numbers s_1, s_2, s_3 are different, or all numbers s_1, s_2, s_3 are the same.

Educational Codeforces Round 182 (Rated for Div. 2)

Contest is running

01:59:35

Contestant



→ Submit?

Language: Python 3.13.2 
 Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Submit

Input

The first line contains a single integer t ($1 \leq t \leq 1000$) — the number of test cases.

Each test case consists of two lines:

- the first line contains a single integer n ($3 \leq n \leq 40$);
- the second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 40$).

Output

For each test case, if a suitable pair of integers l and r ($1 \leq l < r < n$) exists, output these two integers (if there are multiple suitable pairs, you can output any of them). Otherwise, output two integers equal to 0.

Example

input	Copy
4 6 1 2 3 4 5 6 4 1 3 3 7 3 2 1 0 5 7 2 6 2 4	
output	Copy
3 5 0 0 1 2 2 4	

Note

Consider the examples from the statement:

- in the first example, the array is cut into parts $[1, 2, 3]$, $[4, 5]$, $[6]$; $s_1 = s_2 = s_3 = 0$;
- in the second example, there is no suitable cut;
- in the third example, the array is cut into parts $[2]$, $[1]$, $[0]$; $s_1 = 2$, $s_2 = 1$, $s_3 = 0$;
- in the fourth example, the array is cut into parts $[7, 2]$, $[6, 2]$, $[4]$; $s_1 = 0$, $s_2 = 2$, $s_3 = 1$.



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